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 INTEGRATED REAL-TIME TENT AND EQUIPMENT ALLOCATION
 FRAMEWORK

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ENHANCING EVENT LOGISTICS EFFICIENCY THROUGH AN INTEGRATED
REAL-TIME TENT AND EQUIPMENT ALLOCATION FRAMEWORK

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DECLARATION

I hereby declare that the work in this project is my own except for quotations and citations which have been duly acknowledged.

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ABSTRACT

AR Events is a medium sized wedding and feasting event equipment rental firm based in Jalan Petaling, Jalan Panasir, Bandar Sunway, Petaling Jaya, offering wedding equipment, tables and chairs, cooling systems and fancy fittings for weddings, feast outlet or other functions rentals. With more than one thousand assets under their charge, it has four employees, yet all the records about their bookings and allocations are in a mobile phone list, invoices manually prepared and payment and transport records stored separately. This does not provide a single picture of what is pledged on a particular day, there is no means of assessing the profit of an event and there is no data available to verify the value of the season. In this project, we aim to propose an integrated solution for real-time tent and equipment allocation system for companies of this magnitude. This is supported by literature research on rental inventory management and resource allocation, as well as by collecting the requirements from the owner, to document the existing method. The analysis and the design include how bookings will be captured, available asset data, loss/damage data, reconciliation data, wage settlement data and event cost data will be captured, along with an analytical approach for demand data and an analytical dashboard for profit. One fact influences the design of the machine, equipment is booked out on the previous day of an event and is dismantled on the following day. hence a booking is a 3 day window with stock listings, not just a single day. Availability issues are thus specified as overlapping intervals not matching dates, this is what the allocation logic addresses. Result is that a documented set of requirement specifications, logical models, database design and interface design are created. Implementation and testing delayed to the large project, this report becomes the specification for the large project. The work demonstrates how allocating resources in real-time can transform the traditional business processes.

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CHAPTER 1: INTRODUCTION

1.1 OVERVIEW

The event rental business holds a particular place in the realm of small and medium enterprise. Unlike retail businesses that sell product one time or service firms that do not hold any physical stock, the revenue stream for an event rental company is built on the recurring deployment and retrieval of the same physical assets. Each canopy, table, chair and fan is booked to a customer for a given period, moved to a site, erected, removed, recovered and then made available again. Profitability is consequently less a question of the size of stock held than of the precision with which that stock is tracked, allocated and recovered over overlapping contracts.

AR Events is a medium-sized rental company for event equipment, located at Bandar Sunway, Petaling Jaya, Selangor. The company has its own depot for its equipment and four staff responsible for transport, assembly and disassembly. The inventory includes thirty canopy sets in five size variations, one hundred and twenty eight tables of three types, eight hundred chairs of two types, seven air coolers, thirty fans of two mounting types, twenty extension cables and a range of decorative accoutrements such as LED lights, bulbs, scallops, ceiling spider drapes, table cloths, chair covers and chair ribbons in a variety of colour combinations. The business caters to weddings, kenduri, corporate celebrations, anniversaries, birthdays and

Nichayathartham rituals. It gets bookings through Instagram direct message, WhatsApp, telephone calls, walk-in queries and recommendations from existing customers. With all this asset base, the complete booking and allocation process is done by inputs made into a list on the owner's cell phone. Invoices are created manually, and the owner keeps chat logs, bank transfer records and shipment details separately. There is no record system that connects a booking to the specific assets that were committed to that booking. There is no way to know what is available on a contested date and no data on how much the costs were to be able to evaluate what an individual event was worth.

This project aims to fill that gap by creating an integrated real time tent and equipment distribution structure for the AR Events. The framework is not about replacing the existing customer channels of the owner that perform well, but about providing a framework for the operational and financial data that these channels already create but which is now never collected in a useful form.

1.2 PROBLEM BACKGROUND

Small and medium-sized event rental businesses often start off informally. An operator buys a small supply of canopies and furnishings, services a local customer base mostly by word of mouth, and handles allocation from memory or a scribbled note. This method is fine if stocks are low and the operator is assured of what's out, back, and what's free. If there is no formal in place there is no obvious costs.

There is that informality which are preserved for a long time even after it is ideal. However, the study by Ulupui et al. (2024) revealed that the barrier to the adoption of the formal management practice does not seem to be ignorance of its value levels but rather a problem of implementation given that the majority of micro, small and medium firms operate without formal internal control practice. The findings are also similar for the Malaysian context by Wan Mahdi et al. (2025), which attribute the reluctance to use the digitalization tools among traditional micro-enterprises as a result of resource constraints, entrenched operation working practices, and cultural preferences without questioning the benefits of digitalization.

As the volume of assets increases, it becomes increasingly difficult. Rental operations do not have an allocation difficulty that matches the value of the inventory, but rather the number of potential sets of assets (allocation) that can be used across the concurrent commitments. A business could, for example, have 30 sets of tree canopies in 5 different combinations, 3 types of tables and 2 types of chairs. It is not a single pool of stock but rather a cluster of interconnected pools of stock that will all need to be individually checked against each required time period. Availability even at a level of the individual asset from memory or a linear list is unreliable, and a successful operator must wind up making commitment decisions on limited knowledge, as soon as many events start to overlap. Based on 154 SMEs, Oyetade et al. (2024) found that any form of tracking inventory systems has a statistically significant impact on the performance of the enterprise, irrespective of how the inventory is stored and managed. That is, the right storage and knowing what you have don't replace knowing where you have it stored.

The second and more important difficulty is financial visibility. Revenue each event is easily quantifiable in a rental model as this is on the invoice but the cost per event is amortised across the transport, labour, consumables, cleaning and asset depreciation and replacement. If these expenditures are never recorded against the event that incurred them, the operator sees turnover but not margin. In practice this means that pricing, discounting and investment decisions are made without understanding which kinds of event or items of equipment are genuinely profit generating. Panigrahi et al. (2024) argue that inventory management is a bigger deal for smaller firms because they are more constrained by cash flow, storage capacity and exposure to stockouts than larger firms. An unmeasured margin is therefore a materially greater risk to a business of this size than to a large one.

Labour's third challenge is asset recovery. Loss and damage in an operation where equipment is to be moved, assembled outside, and taken under a time constraint is an inevitability. Detection, ascription and retrieval really matters. If there's merely a rudimentary log of whatever has been passed out, how the reconciliation is accomplished at the disassembly point relies on the memory and diligence of those who show up, and undetected losses become simply vanilla purchases in the asset base. In the study by Tan and Sidhu (2022) on automated identification for supply chain management, accuracy benefits gained from systematic tracking is emphasized with exposures arising from a lack of systematic tracking.

The fourth challenge is related to demand planning. The rental sector in the events

industry of Malaysia is very seasonal; it peaks at the time of wedding seasons and cultural as well as festive seasons. A good operator will feel the seasons he or she can expect to be busy in, but that's a notion, not a set of data. When companies have not implemented advanced information systems, the only information available is what has already happened in the past and forecasting is rarely used and instead relies on the managers' judgment. Without date bookings, there's no way to judge that either the reference is just wrong or it can be used to schedule stock acquisitions or maintenance and evaluate against the potential.

AR Events demonstrates all these circumstances. The business has grown to the point that it coordinates across a list of mobile phones more than a thousand individually rentable assets, financial performance at the level of the individual event is entirely unmeasured, loss and damage are only identified during dismantling, and seasonal demand is only understood as recollection. The firm is not failing as a consequence, but it is functioning blind to the parts of its own performance that most directly affect whether it expands profitably.

1.3 PROBLEM STATEMENT

AR Events operates a rental inventory of over a thousand distinct assets, in a number of canopy size configurations, table kinds, chair types, cooling and lighting equipment and ornamental fittings, using entries listed in a list on the owner's cell phone. Bookings are received through five different channels, bills are generated manually, and records of payment, conversation and transport are stored separately with no unified record linking a booking to the precise assets that

have been allocated to it.

Five unique difficulties arise from this setup.

First, the firm does not have a real-time perspective of asset level availability. The phone list is a list of bookings, not of committed stock, therefore the owner cannot tell exactly what products are uncommitted for a given period. The owner says that, when people have asked, it has been because the dates were actually fully committed and not because availability was in question, and there has always been enough available stock for the job accepted thus far. The problem is, therefore, not one of observable loss of revenue. That availability is appraised by memory in a live interaction with a customer. That a refusal cannot be verified by reference to a record. That the adequacy of stock at any time is a feature of current booking activity, not a permanent state of affairs. Such exposure increases as the business expands.

Secondly, the business has no visibility of cost or profit at any level. The owner said figures for cost and profit are not known. Expenditure is never allocated to the event that caused it, so it is not possible to determine the profitability of a particular booking, or the relative profitability of different types of events or categories of equipment. Pricing and investment decisions are therefore made without financial evidence.

Third, loss and damage are identified reactively. When the unit is disassembled, missing and damaged items are recorded and charged to the customer at market

replacement price, plus the cost of shipping the replacement. However, because the outbound record is an informal listing as opposed to a verified allocation, detection depends upon the diligence of the crew on hand, and there is no cumulative record of which asset categories suffer the most losses.

Fourth, wage liabilities are not tracked. The daily pay rate for workers is adjusted based on the physical demands of the job and is paid per worker per job. There is no record of a worker to the jobs worked and the sums settled, thus outstanding pay liabilities are kept informally, and labour, the largest controllable cost in the operation – is not attributable to the event that gave rise to it.

Fifth, the seasonal demand is unconfirmed. The owner advises February, April, May, June, August, September and December to be busy months and January, March, October and November to be slower months, with wedding seasons driving the sharpest spikes but there is no structured historical record against which these patterns can be confirmed, quantified or used for planning.

AR Events will continue to operate with a large and expanding asset base but will do so without a structured approach to booking, asset allocation, recovery, labour and cost data. It will not be able to determine where its profit is created or where its capacity is being wasted.

1.4 PROJECT AIM

The aim of this project is to propose an integrated real-time tent and equipment allocation framework that provides the owner of AR Events with accurate, asset-level visibility of equipment availability across concurrent bookings, and quantified visibility of the cost, revenue and profit generated by each event.

The project aims to shift the business from a situation where allocation and price decisions are made on the basis of memory to a situation where they are made on the basis of a verifiable record of what is committed, what is available, what each event costs and what each event returns. According to the scope of the minor project, the focus is on analyzing the current operational process, defining the requirements for a suitable system, creating the system architecture, data structure, analytical technique and interface required to provide this visibility. The subsequent significant project is the construction of the working system, and this design forms the basis of such construction.

1.5 PROJECT OBJECTIVES

The aim stated above is decomposed into the following objectives, each addressing a problem identified in Section 1.3.

Objective 1: To understand AR Events' current booking, equipment allocation, labour and record-keeping practices by directly engaging with the business owner, identifying how bookings are received, how equipment is committed and recovered, and how operational and financial records are currently maintained.

Objective 2: Develop a real-time allocation model to maintain asset-level availability in overlapping windows of occupancy and identify the uncommitted amount of each equipment variant for any given period.

Objective 3: Develop a data structure to capture cost, labor and revenue at the individual event level to enable calculation of per-event profit, per-event-type profitability and per-asset-category contribution.

Objective 4: Create an analytical way to find and quantify the seasonal demand patterns using accumulated booking records and create a dashboard that gives the owner visibility into booking details and daily, weekly and monthly profit.

1.6 PROJECT SCOPE

The scope of this minor project is limited to the analysis and design phases of the proposed framework. Scope of Work:

- Requirement collection from the owner of AR Events as the only intended user of the system.
- Document the present process of booking, allocation, recovery, labour and record keeping.
- Specification of functional and non functional needs.
- Specification of hardware, software, personnel, network and data resources
- Logical system models formulation comprising use case diagram and data flow diagrams at context and first level.
- Feasibility study on technical, economic, operational, timetable and legal aspects. 9

- Database design includes an entity-relationship diagram encompassing bookings, customers, equipment classifications and variants, allocations, expenses, labour, payments and loss or damage records.
- Design the system architecture and interfaces for owner-facing screens.
- Define the analytical approach for demand trend identification and profit reporting.

The following are outside the scope of this modest project:

- Coding, implementation and deployment of the working system. This is undertaken in the succeeding big project.
- Unit testing, integration testing and user acceptance testing of developed system.
- Integration with third party payment, accounting or logistical platforms.
Automated scheduling of workers or routing of transport.
- Booking functionality directly to the customer, as bookings will still need to be made via the regular channels of the business.
- Processing payroll The framework monitors the salaries owed and paid but does not release the payout.
- A financial audit or verification of the historical accounts of the business.

The system is scoped for single user application. The business has 4 workers . They are not intended users. No worker-facing interface is planned.

1.7 TARGET USER

The only intended user of the proposed system is Mr Appala Raj, proprietor of AR Events, who oversees bookings, pricing, allocation choices, wage settlement and financial records himself. He has no formal technological experience and runs the business largely from his mobile phone, which holds the current booking list. Therefore the system must be operable on a smartphone, require no formal training and fit into the brief durations available between operational commitments.

The four people hired by the business, who deal with transport, assembly and disassembly, were specifically recognized as not needing access to the system. Nevertheless, the system records their operations as data, including the assets they deploy and recover, the losses and damage they discover while disassembly, and the wages they make, all of which are part of the system's records.

AR Events customers are not users but indirect beneficiaries. They communicate with the business on its existing lines of communication and would feel the impact of the system through quicker and surer confirmation of availability, rather than through any interface of their own.

1.8 PROJECT JUSTIFICATION

The rationale for this project is simply that the amount of value AR Events has to manage and the level of sophistication used to manage those assets is not comparable. A rental stock count of over a thousand enables a rental inventory to be coordinated via a mobile phone list that shows the location, inventory number,

and owner. A rental stock count of more than a thousand items enables a rental inventory to be prepared through a mobile phone list compilation of the owner, location and inventory number of the rental. So any improvement in allocation accuracy, or financial visibility will impact a large asset base and the return on a relatively small system will be large.

The financial aspect is more imminent. A business without knowing how much profit it can make from an event, cannot assess if the pricing is sufficient, if a specific type of events are worth engaging in, or if particular equipment positions make business sense. Setting up the cost and profit visibility per event solves the biggest blind spot in the existing business operation, all without having to collect any additional data that the business actually creates on the daily basis, such as manual invoices, payment details, wage payments, transport details etc.

The project also fits the limitations and restrictions of its environment. The owner states that he will be willing to manually record each new booking, about one minute per entry, rather than dealing with a system or manual process that he wants but does not have. Some stored archival data is achieved as part of a manual archive of invoices, a history of chat conversations, and a history of bank transfers, and this data, which can be already present in archives, is added to existing historical data. Existing historic data, in the form of manual invoices, history of chat conversations and bank transfer history are supplemented by existing data.

to meet user requirements in an actual real business versus being based on a generic business, collecting user requirements from an actual user rather than based on a generic template. It is directly connected to real-time resource allocation, relational data modelling and descriptive analytics, and yields a design that is of adequate definition for implementation in the upcoming major project.

1.9 PROJECT TIMELINE

The project is executed across the semester in accordance with the schedule issued by the programme. The full schedule is presented as a Gantt chart in Figure 1.1.

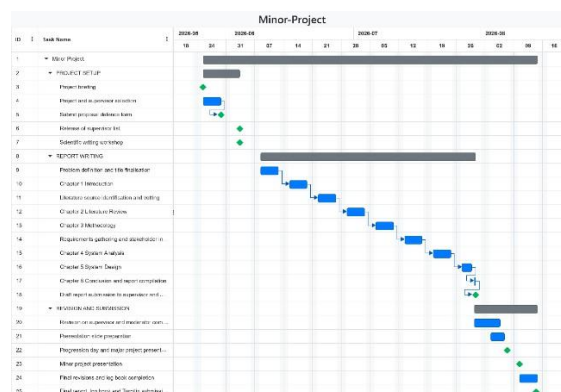


Figure 1 : Project Gantt chart

Literature Review and methodology were written before the gathering of requirements and System Analysis was written before the system design, followed by the latter before the system implementation, thus ensuring a logical progression of each chapter.

CHAPTER 2: LITERATURE REVIEW

2.1 INTRODUCTION

Theoretically and practically, the framework proposed in this chapter is founded on this. The review follows the organization of the four problems of Section 1.3: lack of indicator of asset level availability, lack of cost and profit measurement at event level, reactive detection of loss and damage, and unconfirmed seasonal demand.

There are four stages in the chapter. Section 2.2 analyzes the theoretical theories from inventory control of small businesses, real-time resource allocation segmentation, seasonal demand forecasting, business intelligence for smaller businesses, and asset tracking technologies. Section 2.3 reviews the commercial rental management systems already operating in this market and compares their capabilities against the requirements of a business of AR Events' scale. Section 2.4 synthesises these strands to identify the specific research gap this project occupies, and Section 2.5 summarises the chapter.

Sources were located through Google Scholar, MDPI, ScienceDirect and DOAJ, using keyword combinations covering inventory management, SME performance, resource allocation, demand seasonality, business intelligence and asset tracking. Preference was given to publications between 2020 and 2026, to open-access material that could be read in full rather than in abstract, and where possible to research conducted in Malaysian or comparable Southeast Asian contexts.

2.2 THEORETICAL FOUNDATIONS

2.2.1 INVENTORY MANAGEMENT AND CONTROL IN SMALL ENTERPRISES

The foundational premise of this project is that how a small business tracks its stock materially affects how that business performs. This is not self-evident, since a small operator may reasonably argue that informal methods are adequate at small scale, and the literature has tested the claim directly.

Oyetade et al. (2024) surveyed 154 small and medium enterprises in Lagos State using a structured questionnaire administered to owners and senior staff, examining three distinct dimensions of inventory practice: control techniques, storage systems, and tracking systems. A significant relationship was found in all three. The result of this project is that tracking and control are separated, in that AR Events has a workable control; he knows what he has in the form of a control, whereas he has absolutely no system of tracking and thus no tracking in any form. Among the SMEs, the inventory control is recommended to be made formal at operational level and not informal, recommends Oyetade et al.

A systematic review, limited down to 95 papers for detailed analysis from 175 candidates papers, came to a comparable conclusion by Panigrahi et al. (2024). Their review acknowledges that effective inventory management is particularly relevant for SMEs, because they have less rigorous cash-flow positions, smaller inventories and lower levels of exposure to inventory shortages than large firms.

They also provide an introduction to "know-how" and differentiate between technical and technical and behavioural aspects and suggest a concept for practice of inventory based on the operator's capability rather than the system itself. The project has direct implications for the framework exceeding the owner's technical comfort: whatever the better-designed the framework is, if the owner is not technically comfortable with it, he will not use it.

Pearse et al. (2025) furthered this cause of study to the retail section and identified practices within the supply chain, such as on-time delivery, perfect order fulfilment and stock holding as having a significant impact on the performance of retailers. The notion that an order is performed correctly only when all items in the product line are delivered, work properly and retrieved is pertinent to the field of event rentals.

At the other end of the spectrum, Ulupui et al., (2024) give the explanation behind the success of small enterprises in staying informal. The study targeted 140 micro, small and medium businesses (MSMEs) in Indonesia, specifically those that do not have an internal control system, and revealed that the level of adoption of formal management practice is still quite behind its value. Doing the business of raising awareness is not the obstacle facing AR Events, it's quite likely the business of actually entering the data into the system.

2.2.2 REAL-TIME RESOURCE ALLOCATION AND DECISION SUPPORT

The allocation of rents differs from the allocation of retail inventories in a particular way of significance for the design of the system. Retail stocks are used up once,

rental stocks are paid-off to a period of time, recovered and offered again. Availability, therefore, is not just a number but depends upon the time, and a system has to be able to answer not ‘howManyChairs exist’ but ‘howManyChairs areUncommitted on 14-JUN-2003’.

Teerasoponpong, Sopadang (2022) modeled a decision support system for adaptive sourcing and inventory management that is specifically designed for small and medium scale enterprises (SMEs), and closely resembles the model proposed here. They argue that SMEs require a decision support more appropriate to their resource limitations and not scaled down versions of enterprise systems. The difference between their solution and this one is the domain their solution related to the replenishment of products that must never be out of stock, the event rental business is about recouping products. The allocation logic that is needed is, for that reason, different, as a physical asset needs to be modeled as unavailable for a certain period, and thus be available to use again.

Naturally treating the combined logistic problems of inventory management and transportation planning rather than two independent logistic problems would result in better outcomes, and this issue was explored by Kmiecik (2022). This pertains to AR Events – where not only assets, but also their movement to a site, set up, takedown and return to the site, is committed to. Transport records are currently stored separately to booking records.

An adjacency makes an invaluable implementation precedent to Agpalo et al. (2025). Their Web-based event management and attendance monitor system was installed in secondary schools in rural Philippines to address the fragmented

workflow that involves the use of multiple manual processes, and was validated by ISO 25010 and TAM perspective. Twenty information technology specialists assessed the high reliability level for the operational parameters, while evaluation, involving thirty stakeholders, revealed a high level of agreement between usefulness perception and behavioral intention to adopt. An attitude from their work that carries over to this project is the way they evaluate the system by how well it is accepted by the users as opposed to just technical performance. Another is the fact that replacing an analog workflow with a digital alternative was found to be successful when there is little extra effort required at the point of data capture.

2.2.3 SEASONAL DEMAND FORECASTING WITH LIMITED DATA

The four problems delineated in Chapter 1 relate to unverified seasonality, and in this respect the literature has most directly applicable method.

Borucka (2023) addresses just this scenario in which AR Events finds itself. The study notes that companies that cannot use a modern system of information simply have to rely on past sales as their source of information, and in these cases it is frequently preferred to abandon forecasting rather than the use of a judgement approach. On the contrary, the paper provides comparison between seasonal ARIMA and ARIMA with Fourier terms, exponential smoothing (ETS) and TBATS models for monthly sales data with strong seasonality and underlying trend. The key conclusion is that there is no definitive relationship between model complexity and the accuracy of forecasts and that simple analytical time series methods can be as accurate as machine learning methods with ease of interpretability for a non-engineering decisionmaker. Borucka is emphatic that 18

such procedures can be applied in any enterprise, irrespective of the computerisation level: only a simple series of past observations is required.

This finding is reflected in two aspects of the analytical approach to postulate for AR Events. It first sets the stage to test owners' memories of busy and slow months in relation to the data and not against a large data set or sophisticated modelling systems. Second comes to the question of justifying the choice of a method that is easily understood rather than a more complicated one because if it cannot be understood then it isn't a forecast the owner will act on.

The last approach is from a nod to the supply side where Tadayonrad and Ndiaye (2023) have looked at the issue of seasonality from an inventory perspective, and developed a model of key performance indicators suitable for choosing the proper service level based on the results of the supply reliability factor calculations and the seasonality indexes, where they tested their model against a real dataset. They are the people who have given this project a new idea, the idea of a seasonality index – a number that could be put on unto a month and be used to make decisions regarding stock, maintenance schedules, personnel etc.

2.2.4 BUSINESS INTELLIGENCE AND DASHBOARDS IN SMALLER FIRMS

The last theory strand is related to how output of the analysis should be communicated to a non-technical owner.

Gonçalves et al. (2023) created dashboards powered with Power BI, based upon

the methodology introduced by Vercellis, aimed at the adoption of a range of sales KPIs, and indicated that 22% of SMEs are still not using business intelligence. Their research shows that a single repository of the data from multiple sources allows for concise graphical analysis and very quickly interpretable and that the use of a dashboard is not about too much analytics, but of its timeliness.

Salisu et al. (2021) did a systematic review of BIS adoption literature in the SMEs period 2011-2020 from a CE, technological, organisational and environmental viewpoints. An observation from their review which dovetails with this project's ambitions should dampen down any entrepreneurial dreams in this regard: a lot of business intelligence projects fail to deliver the results they presume they should be getting and even less so in small and medium enterprises (SMEs). It is far more believable to accept a design that has taken into account this danger than to assume it will be accepted due to its usefulness.

Based on the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT), Alsibhawi, et al. (2023) acknowledged six determinants, including change management, knowledge sharing, information quality, IT project management, perceived usefulness and perceived ease of adoption, for business intelligence adoption among SMEs. Both information quality and perceived ease of adoption are directly controllable by a system designer, and are covered by the design decisions given in Chapter 5.

2.2.5 ASSET TRACKING TECHNOLOGIES

Tan and Sidhu (2022) examined the applications of radio frequency identification and IoT in supply chain management and it described the processes involved in using automatic identification to improve inventory control and lessen manual counting. Their review is included here since it represents a high watermark for the benefits asset-tracking can provide – and the opportunities it's being sacrificed by the community.

The AR Events conclusion is negative yet significant. The technologies of "tagging" bring their benefits when the individual asset is valuable, but not numerous. The number of 800 chairs and 128 tables, which are used outdoors and disassembled and assembled by experts who do not use the tag, is an unfavourable case: reader hardware would have to be operated by non-users of the system and tag survival in the field is uncertain. The framework set out in this project therefore is in line with the category-level quantity tracking, introduced instead of unit level identification, which is well justified in chapter 4 and not accidental.

2.2 REVIEW OF EXISTING SYSTEMS

It will be required to conduct an investigation of the current systems related to the project topic and make comparisons between them by this Report Writing and Formatting Guideline. A mature commercial market exists for rental management software, and the framework proposed here must be justified against it rather than presented as though no alternative existed. The following platforms

were reviewed from vendor documentation; they are commercial products rather than academic sources and are cited as such.

Table 1 : Review of Existing System

SYSTEM	PRIMARY MARKET	ALLOCATION HANDLING	FINANCIAL REPORTING	FIT FOR AR EVENTS
Booqable	Small and medium rental businesses	Inventory scheduling with online booking intake	Order-level invoicing	Closest in scale, but built around customer self-service booking, which AR Events does not use
Rentman	Event and AV production	Category and serialised tracking, ABC analysis	Project-level costing	Strong functionally, priced and scoped for production companies
Flex Rental Solutions	Professional AV and live events	Serialised assets with optional RFID	Rental reporting and utilisation	Substantially exceeds requirements; assumes warehouse staff as system users
TapGoods	Party and event rental	Availability updates automatically on order creation, check-in and check-out	Order-level revenue	Functionally well matched, but subscription cost is disproportionate at this scale
EZRentOut	General equipment rental	Real-time availability, location and condition, field damage reporting	Utilisation and revenue reporting	Damage reporting is directly relevant; requires field staff participation
Alert Rental	Party, event, tool and heavy equipment	Multi-route asset tracking with staff assignment	Contract and billing management	Designed for multi-branch operations

Three observations follow from this comparison, and together they define the position this project occupies.

First, every platform reviewed solves the asset-level availability problem competently. The technical problem of knowing what is committed on a give date is not an open research question, and this project does not claim to solve it novelly.

Second, all of these platforms assume an operating model that AR Events does not have. They assume multiple system users including warehouse and field staff, customer-facing booking intake, and an existing cost structure that the business can enter into the system. AR Events has one system user, receives bookings through five conversational channels that the owner has no wish to replace, and does not currently know its own costs. A platform that requires cost data as an input cannot help a business whose central problem is that it has never captured cost data.

Third, and most significantly, none of these platforms treats the derivation of cost structure as part of the problem. They are built for operators who already know their unit costs and want to apply them efficiently. The framework proposed here inverts that assumption, treating the capture and construction of a cost baseline as the first function of the system rather than a prerequisite for using it.

2.3 SYNTHESIS AND RESEARCH GAP

Drawing the strands together, the literature establishes four positions relevant to this project.

Inventory tracking practice measurably affects small enterprise performance, and tracking is separable from storage and control as a determinant of that performance (Oyetade et al., 2024; Panigrahi et al., 2024). While there are studies that explore how to make the replenishment decisions for services feasible in a small business context, they don't consider rental cycles and recovery (Teerasoponpong & Sopadang, 2022; Kmiecik, 2022). Small historical observations can be leveraged to detect and measure seasonal patterns, without needing any sophisticated infrastructure or complex analysis techniques (Borucka, 2023; Tadayonrad & Ndiaye, 2023). Smaller companies are slower to adopt business intelligence that comes in the form of a dashboard, but it can help non-technical decision makers.

Most projects do not deliver in full due to EoA being treated as a feature more than a design consideration (Gonçalves et al., 2023) and others fail to address barrier issues (Salisu et al., 2021; Alsibhawi et al., 2023).

Set against the Malaysian context, Wan Mahdi et al. (2025) observe that traditional micro, small and medium enterprises continue to operate manually despite clear potential benefits, attributing this to resource limitations, entrenched manual practice and cultural norms, and applying the Technology Acceptance Model extended with trust to explain the resistance. Azuar and Nehru (2024) report that 21,591 Malaysian SMEs had adopted digitalisation through the MDEC 100 Go Digital initiative across 2023 and 2024, and identify finance as the

principal barrier to further adoption. Both sources support the conclusion that a framework for AR Events must be inexpensive and must impose minimal behavioural change to have any prospect of being used.

The gap this project occupies can now be stated precisely. The academic literature addresses inventory tracking in SMEs, decision support for SME sourcing, seasonal forecasting under data scarcity, and business intelligence adoption, but does not address these as an integrated problem within a recovery-based rental operation. The commercial literature addresses recovery-based rental operations comprehensively, but assumes a scale, a staffing model and a pre-existing cost structure that a four-worker business coordinating bookings through a phone list does not possess.

Neither body of work addresses the specific case of an asset-intensive micro-enterprise that must simultaneously establish asset-level allocation visibility and construct a cost baseline from scratch, using data captured by a single non-technical user at a rate of approximately one minute per booking. That is the gap this framework is designed to occupy.

2.4 CHAPTER SUMMARY

This chapter reviewed the theoretical foundations, existing systems and related research bearing on the proposed framework. Section 2.2 established that inventory tracking practice affects small enterprise performance, that decision

support can be designed within SME constraints, that seasonality can be quantified from limited historical data using interpretable methods, and that dashboard adoption depends more on ease of use than on analytical power. Section 2.3 compared six commercial rental management platforms and found that while all solve asset availability competently, all assume an operating model, staffing structure and cost baseline that AR Events does not have. Section 2.4 identified the resulting gap: the integration of allocation visibility with cost baseline construction for a single-user, asset-intensive micro-enterprise.

Chapter 3 sets out the methodology through which this framework is developed, including the selection of a development model, the pre-development, development and post-development phases, and the data collection techniques used to gather requirements from the business owner.

CHAPTER 3: METHODOLOGY

3.1 INTRODUCTION

In this chapter the research process and techniques adopted in the development of the proposed framework for real-time tent and equipment allocation are discussed. Chapter 1 established what the project aims to achieve, chapter 2 established what is already known, this chapter establishes how the work was carried out and why those particular methods were chosen.

The chapter first sets out the overall research design and its rationale, then justifies the selection of a system development methodology against the alternatives considered. It then describes the pre-development, development and post-development phases in accordance with the requirements of the Report Writing and Formatting Guideline and concludes by outlining the data collection and analysis techniques used. The project schedule is shown in Figure 1.1.

3.2 RESEARCH DESIGN

This project uses a qualitative, single-case study design with a design science orientation. Each item in that description has a specific justification.

The study is qualitative because the problem it seeks to solve is not a matter of measuring a known quantity but understanding the undocumented process. Before this project, no one had written down how AR Events takes a booking, commits

equipment, recovers it, settles wages, or accounts for what an event costs. Before any system could be specified, that process had to be elicited and documented. Eliciting an undocumented working practice is inherently qualitative work.

The study is a single-case design, since the framework is developed for a particular business, with the design determined by the constraints of that business. A multi-case or survey approach would yield generalizable findings about event rental businesses in aggregate, but would not yield a specification detailed enough to work from. The trade-off is deliberate: it is a trade-off that favors depth rather than breadth, and the limits to generalizability that this imposes are discussed in Chapter 6.

The study is design science in orientation in that its output is an artifact a specified framework rather than a proposition tested against evidence. The research question is not “is this true” but “can a workable framework be designed within these constraints”, and the framework itself is the answer.

This approach is consistent with the precedent set by Teerasoponpong and Sopadang (2022) who developed a decision support system for small and medium enterprises by first characterizing the constraints those enterprises operate under and then designing to them rather than downwards adapting an enterprise scale solution.

3.3 SELECTION OF SYSTEM DEVELOPMENT METHODOLOGY

Three development methodologies were considered.

The concept of Agile was looked at and rejected. The principle behind the value of Agile is the delivery of working software to users who provide feedback on the requirements, and assumes a development time long enough to allow multiple iterations. This is not the case here. The scope of the minor project is limited to design, thus no working software is available for the user to act upon, and only one stakeholder instead of a product team. Doing Agile would be taking a methodology that has a mechanism it never uses throughout the project.

The idea of prototyping was taken into consideration and rejected partially. Prototyping is appropriate for clarifying requirements a stakeholder can't express abstractly and interface mockups are indeed a part of the deliverable in Chapter 5. During this project the need for a per-worker wage settlement was identified through the labour screens, which was not obtained from the initial interview (as described in Section 4.2). Prototyping as a whole-project approach, however, assumes a lot of uncertainty in the requirements, since the requirements here were found to be rather consistent once elicited. Prototyping is thus not the main methodology, but rather a technique used during the design phase.

Waterfall was selected. The rationale is based on three attributes of this project.

First, they are all requirements from one stakeholder and once collected, they are

relatively fixed, which negates the main advantage for an iterative approach. Secondly, project deliverables are sequential: system analysis must be finished before the requirements can be documented, and system design must be completed before the analysis is completed. Third, the academic assessment structure is sequential by itself, a chapter is submitted in order and builds on the previous one.

Inflexibility when requirements change, and the fact that errors can only be identified late in the game, is a very real problem that was faced in this project twice with Waterfall. The finding of the occupancy window required a change in requirements, which were already documented, and the requirement for wage settlement only came during interface design. Both are not hidden in any way, but are recorded in Section 4.2. There were two aspects to mitigation: Firstly the requirements were not assumed to be correct but validated with the stakeholder after documenting the requirement; Secondly the design was checked against the requirements before the major project started to implement the design. The restriction is still admittedly in Chapter 6.

This is indirectly supported by Panigrahi et al. (2024) who concluded that inventory practice in small firms is as much affected by the technical and technological and behavioral know-how of the operator as in the system adopted. When the result is so dependent on the ability of one person it is better to have a methodology that involves understanding of the user up front than defer it until later iterations.

3.4 PRE-DEVELOPMENT PHASE

The pre-development phase established the problem, the knowledge base and the requirements.

Problem identification: The issue was determined based on first-hand experience working at AR Events and an interview with the business owner. It's an open secret, as it gave both an advantage: being familiar with the operation made the elicitation process quicker but also a risk of assuming knowledge instead of verifying it. This risk was reduced by making all operating assumptions explicit and not assuming on the owner's memory.

Literature review: The keywords were used in a structured search of Google Scholar, MDPI, ScienceDirect, and DOAJ. The combinations of the keywords covered Inventory Management, SME performance, real-time resource allocation, seasonal demand forecasting, business intelligence adoption, and asset tracking. The following criteria were used to screen the candidate sources: (1) publication date 2020–2026; (2) availability of the full text; and (3) the source's relevance to small-enterprise situations, rather than enterprise-scale situations. There were 15 which met all three and these are the basis of Chapter 2.

Requirements gathering: The requirements were obtained by conducting a structured interview with Mr Appala Raj, who is the owner of AR Events on 02 July 2026 on the basis of a question guide that was prepared. The guide contained twelve sections: bookings volumes and event types, the make-up of stock, booking channels, existing tracking system, capacity – and turned-away bookings, loss and

damage, understanding of cost and profit, availability of historical records, seasonal trends, owner's information priorities, workforce and system access, and intention to record bookings in the future. Equipment occupancy window was covered in the next round with a further eight areas, which included pricing basis, delivery charging, labour cost, transport assets, payment terms, loss and damage pricing, and partial fulfilment.

Review of existing systems: The operating constraints defined in Section 2.3, and based on the documentation provided by the vendors, were used for comparison and evaluation of six commercial rental management platforms.

3.5 DEVELOPMENT PHASE

The development phase for this minor project covers system analysis and system design; construction is deferred to the major project.

System analysis. Interview responses were formulated into functional and non-functional requirements and each requirement is traced back to a specific statement provided by the owner and no requirement is left on assumption. Hardware, software, people, network and data resources were specified. Use case diagram, data flow diagram (context level), data flow diagram (first level) were created. The feasibility was evaluated for technical, economic, operational, schedule and legal aspects. This work is presented in chapter 4.

System design. The design phase results in the structure of the database, the

architecture of the system and the design of the interface. The entity-relationship diagram includes the following entities: bookings, customers, equipment category, equipment variants, allocation, costs, labour, payment and loss and damage records; and cardinality is based on the operational fact that a bookings commits several equipment variants, and that an equipment variant is committed by many bookings throughout time.

The analytical part is intended, but not carried out. An interpretable time series approach (as detailed in Borucka (2023)) is specified rather than a machine learning approach, based on the evidence that more complicated models do not necessarily produce more accurate outlooks for seasonal forecasts, and that whether a non-specialist uses a forecast has a significant effect on whether it is interpreted. As in Tadayonrad and Ndiaye (2023), seasonality is incorporated as an operational index for each month, and not as a descriptive observation, and can be used to inform decisions relating to stock and scheduling.

The dashboard design is based on the premise set forth by Gonçalves et al. (2023) that a dashboard's value is in how quickly a decision maker can read it and not how sophisticated the analytics are. In the Small Businesses' BI adoption research, Alsibhawi et al. (2023) determine that business intelligence uptake depends on perceived ease of adoption and information quality, which is important in making sure that decisions are made that are easier to adopt.

Design validation. The documented requirements and the proposed design were then discussed with the owner to ensure that the system specified solves the problems

identified and the data entry requirement is acceptable.

3.6 POST-DEVELOPMENT PHASE

The post development is here defined as specification, since implementation is not part of the scope of the minor project. There are three activities that are outlined for the major project.

Testing strategy. Unit and integration testing will be used to validate allocation logic, especially when a booking's requested quantity matches the available quantity, and when the same variant is booked for two consecutive bookings with a minimum recovery time between each booking is less than the length of the booking.

User acceptance testing. Technical criteria and user acceptance constructs will be assessed for acceptance evaluation, as done by Agpalo et al. (2025), because a system that satisfies all technical criteria may still be rejected if the single intended user discontinues the use of the system.

Evaluation criteria. The framework will be tested for the ability of the owner to know how much asset he is available for a contested period without having to look at his phone list, whether per-event profit is calculable for a completed booking and whether the seasonal trend found by adding the profits together matches his statement of the busy and slow months. The third is the one in which the data is not in accordance with the owner's memory, but rather that is a finding, not a failure.

3.7 DATA COLLECTION AND ANALYSIS TECHNIQUES

The primary data were collected by a structured interview with the business owner designed with the question guide described in section 3.4. A decision was made to use the interview method rather than questionnaires, or surveys, because the population of interest, being one, made the use of statistical instruments inappropriate, and because the aim was to elicit an undocumented process rather than to measure attitudes within a sample. This was supplementary and observations were available as a result of previous experience in the business.

Secondary data consists of the business's existing documentation such as manually created invoices, message history within the booking channels, bank transfer records and transport information. These are not consolidated and are the property of the owner. The model is an important part of this project because they provide a reconstructable historical dataset, and thus the seasonal analysis described in Chapter 5 do not have to wait for data to accrue from zero. As of the date of this writing, one invoice had been secured, and another one to two months of additional records were anticipated; the impact of this limited number of records is discussed in Section 5.6.2.

Analysis technique. Interviews data was analyzed using thematic categorisation, and responses grouped against the problem areas identified in Section 1.3 and then each identified requirement traced back to the problem statement that led to it. Assumptions made prior to the interview were changed when a response challenged their validity, and relevant parts of Chapters 1 and 2 were revised.

Ethical considerations. The owner of AR Events provided permission for the study and naming of the business in this report. No personal data from customers was gathered, no financial account information was accessed and no worker was interviewed or observed for research. No worker names are included in this report only label names. Commercial sensitive information is not reported at a detailed level, but rather in terms of the operational structure.

3.8 CHAPTER SUMMARY

This chapter found that the project was of a qualitative, single case type with a design science orientation and that the Waterfall model was chosen over Agile and Prototyping due to single stakeholders' requirements, sequential dependent deliverables, and fit within academic reporting structure. The pre-development phase includes the problem identification and literature review, requirements gathering (two readings), and existing system review, while the development phase includes the system analysis and design (construction is deferred), and the post-development phase specifies the approach to testing, acceptance and evaluation of the major project. Data collection has been done with structured stakeholder interview and reconstructable secondary records using thematic categorisation with requirement traceability.

In Chapter 4, the system analysis developed using the methodology is presented, which involves the requirements documented, resource specifications, use case model, data flow diagrams and feasibility assessment.

CHAPTER 4: SYSTEM ANALYSIS AND DESIGN

4.1 INTRODUCTION

This chapter provides the specifications of the proposed real-time tent/allocation of the equipment system. It summarizes the results of the requirements gathering project outlined in Chapter 3, converts the results into functional and non-functional requirements, identifies the resources the system needs, such as hardware, software, people, network and data resources, creates the logical models of the system, and evaluates the system feasibility along five dimensions.

In this chapter, the organisation is by traceability. Each functional requirement in Section 4.3 is mapped to a response obtained during the requirements gathering process, thus eliminating any requirement on assumption or what a system of this type typically contains. If a requirement has been deduced and not explicitly written, this is marked accordingly.

4.2 REQUIREMENTS GATHERING SUMMARY

The requirements were obtained through a three-step process: a structured interview involving 12 areas, a follow-up with eight additional questions that uncovered missing requirements during analysis, and a final requirement that came up during interface design. The following summarizes the responses; they provide the evidence for all the requirements listed in this chapter.

Table 2 : Requirements Gathering Summary

NO	AREA	RESPONCE
1	Booking volume and event types	No fixed monthly volume; depends entirely on incoming orders. Event types: weddings, kenduri, corporate functions, anniversaries, birthdays, Nichayathartham
2	Inventory composition	30 canopy sets (5 transparent, 25 normal) across five size configurations; 128 tables of three types; 800 chairs of two types; 7 air coolers; 30 fans of two mounting types; 20 extension cables; LED lights, bulbs, scallops, ceiling spider drapes, table cloths, chair covers and chair ribbons in multiple colours
3	Booking channels	Instagram direct message, WhatsApp, telephone, walk-in, and referral from existing customers; proportions vary
4	Current tracking method	Entries recorded in a list on the owner's mobile phone
5	Capacity and refused bookings	Bookings have been declined because dates were fully committed, not because availability was unclear; occurs occasionally
6	Loss and damage	Tracked via the phone list; losses and damage identified during dismantling; customer is charged for lost or damaged items
7	Cost and profit knowledge	Not known
8	Historical records	Manually prepared invoices; chat histories; bank transfer records; transport details — all held by the business owner
9	Seasonality	Busy: February, April, May, June, August, September, December. Slower: January, March, October, November. Wedding seasons produce the sharpest spikes
10	Information priority	Booking detail, and daily profit detail
11	Workforce and access	Four workers; workers do not require system access
12	Data entry commitment	Willing to record each new booking personally, approximately one minute per entry
13	Equipment occupancy window	Equipment is dispatched one day before the event and dismantled the day after; each booking therefore occupies stock for approximately three days
14	Pricing basis	A price list exists; pricing starts from RM250 and scales with the number of canopies, tables and chairs booked
15	Delivery charging	Delivery is provided free of charge within Selangor and is not billed as a separate line
16	Labour cost	Workers are paid RM50 per day, rising to RM70 for heavier work
17	Transport assets	The business owns a lorry and a Hilux; transport is performed in-

NO	AREA	RESPONSE
		house rather than hired
18	Payment terms	Half the price is taken as a deposit on booking; the balance is sometimes settled after the event; late payment occurs
19	Loss and damage pricing	Charged at market replacement price plus the transport cost of obtaining the replacement — a chair with a market price of RM5 is charged at RM7
20	Substitution practice	Where a requested canopy size is unavailable, an alternative size is offered with its revised price. The customer either accepts, cancels, or the order is fulfilled using stock sub-rented from a partner event company
21	Partial fulfilment	Has never occurred; available stock has always been sufficient for accepted bookings
22	Wage settlement	Wages are settled per worker per job. Days worked and the applicable daily rate are recorded for each individual worker, and wages can remain outstanding after a job is complete

Five findings from this exercise materially shaped the design and are stated explicitly.

The first is that there is no visibility of cost and profit at any level in the business (response 7). This makes financial measurement a key feature of the system and sets this aside from the platforms discussed in Section 2.3 that assume cost data as an input.

Second, the availability requirement is meant to be justified on different grounds than was intended by responding to both 5 and 21 separately. Unclear availability had been assumed to be the reason for the loss of bookings on the business. Response 5 has proved that there were refusals due to dates being actually full, and response 21 has proved that there has never been any partial fulfilment because there has always been sufficient stock for accepted work. Checking availability should not, therefore, be justified as revenue recovery. There are three

reasons for this: First, it eliminates the need to search through a phone directory during a real customer call; second, it allows a refusal to be backed up by a fact sufficiency of inventory at this time, but not an inherent characteristic of the business; and third, it makes it easier to refuse a customer who requires a lower inventory. This has been updated in chapters 1 and 2.

Third, response 13 is a basic design of the data model. Equipment is booked the day before an event and is taken down the day after, so a booking does not actually 'take up' stock on one day, but on a spanned 3 day period. Availability is then determined by overlapping time segments, and two events occurring on two different days will compete for availability if they have the same event date. An initial version of FR-04 and FR-05 modeled availability based on a single date and was wrong; both were updated and added FR-04a to get window automatically.

Fourthly, responses 14-19 provide the cost and pricing structure that the owner said he did not know. It is important to note that the owner has no knowledge of his profit, but the ingredients needed to determine it are available. The direct, calculable costs of labour are at RM50–RM70 per worker/day over 3 days. Transport is carried out using the owned assets, making it an internal cost that can be recovered by estimating the fuel and maintenance costs, not by an amount on the invoice. Revenue is given by a known price scale based on the number of items. Loss and damage recovery is based on explicit formula market price plus replacement transport. The result is that the profit for each event can be calculated

with data that the business already produces, which is the key to the viability of FR-10.

Fifth, response 22 does not appear until the design process of the interface, and the answer is revealed as such. Response 16 set the daily rates but not the settlement process. When designing the labour screens, it was realised that it would not be possible to have a single labour total for a booking, without knowing who had actually received payment, and who had not. When asked, the owner verified wages are settled on a case-by-case basis and can still be outstanding after a job. This necessitated a one-to-one transformation in Chapter 5 from a single aggregate labour entity to per-worker arrangement in that chapter and the introduction of FR-20. The episode is not a failure of the interview, but is a weakness of the Waterfall model (Section 3.3), as it wasn't known until the interface exposed that requirement.

Lastly, response 12 makes a hard design constraint. The caps on interaction cost are around one minute per booking entry, which is the limit a design can get and still be viable in a working week.

4.3 FUNCTIONAL REQUIREMENTS

Table 3 : Functional Requirements

ID	REQUIREMENT	TRACED TO
FR-01	The system shall maintain a register of all equipment variants with owned quantities, including size, material and colour variants	Response 2
FR-02	The system shall record a booking with event date, event type, customer, and the equipment variants and quantities committed	Responses 1, 2

FR-03	The system shall record the channel through which each booking was received	Response 3
FR-04	The system shall calculate and display uncommitted quantity per equipment variant across a specified occupancy window, not a single date	Responses 4, 5, 13
FR-04a	The system shall derive each booking's occupancy window automatically as the day before the event through the day after it, while permitting manual override	Response 13
FR-05	The system shall prevent a booking from committing a quantity exceeding uncommitted availability at any point within its occupancy window	Responses 5, 13, 21
FR-06	The system shall record equipment returned after each event and identify discrepancies against quantities dispatched	Response 6
FR-07	The system shall record loss and damage against a booking, with the amount charged to the customer	Response 6
FR-08	The system shall record costs incurred per event, categorized by transport, labour, consumables, sub-rental and replacement	Responses 7, 16, 17, 20
FR-09	The system shall record the deposit taken, the balance outstanding and the payment status of each booking, including overdue balances	Responses 8, 18
FR-10	The system shall calculate profit per event as revenue less recorded costs	Responses 7, 10
FR-11	The system shall support entry of historical bookings reconstructed from existing invoices and records	Response 8
FR-12	The system shall generate a monthly demand measure per equipment category from accumulated booking records	Response 9
FR-13	The system shall present a dashboard showing booking detail alongside daily, weekly and monthly profit	Response 10
FR-14	The system shall produce an invoice for a booking from recorded booking and pricing data	Response 8 (derived)
FR-15	The system shall report profitability aggregated by event type and by equipment category	Responses 7, 10 (derived)
FR-16	The system shall maintain a price list and quote a booking from the quantities of canopies, tables and chairs committed	Responses 14, 15
FR-17	The system shall calculate a loss or damage charge as market replacement price plus replacement transport cost	Response 19
FR-18	The system shall flag bookings with an outstanding balance after the event date	Response 18
FR-19	Where a requested variant is unavailable, the system shall identify eligible substitute variants with the resulting price difference, and record the outcome of the offer including fulfilment from sub-rented external stock	Response 20
FR-20	The system shall record wages payable per worker per booking, comprising days worked, applicable daily rate and settlement status	Response 22

FR-14 and FR-15 are marked as derived because they were not requested directly.

FR-14 follows from the existence of manual invoicing as an unautomated task the system is already capturing the data for. FR-15 follows from the stated need for profit visibility: knowing total profit without knowing which event types or equipment categories generate it would leave the pricing and investment decisions identified in Section 1.2 no better informed than before.

4.4 NON-FUNCTIONAL REQUIREMENTS

Table 4 : Non-Functional Requirements

ID	REQUIREMENT	JUSTIFICATION
NFR-01	Booking entry shall be completable in approximately 60 seconds	Response 12 sets the interaction budget
NFR-02	The system shall be operable on a smartphone	Response 4; current records are kept on the owner's phone
NFR-03	The system shall be usable without formal training	Sole user has no formal technical training
NFR-04	Availability queries shall return within 3 seconds	Availability is checked during live conversation with a customer
NFR-05	The system shall operate on intermittent mobile connectivity	Bookings are handled away from a fixed premises
NFR-06	Booking and financial data shall be recoverable following device loss	The phone currently constitutes a single point of failure for the entire business record
NFR-07	Access shall be restricted to the authenticated owner	Response 11; single-user system holding commercial financial data
NFR-08	Analytical output shall be interpretable without statistical training	Borucka (2023); Gonçalves et al. (2023)
NFR-09	Interface language shall be English	Owner's stated preference

NFR-06 deserves comment. The requirements exercise did not raise data loss as a concern, but the current arrangement places the entire operational and financial history of the business in a single mobile phone list with no stated backup. Identifying this is a legitimate function of system analysis, and the requirement is included on the analyst's judgement rather than at the stakeholder's request.

4.5 SYSTEM RESOURCE SPECIFICATIONS

Hardware resources. Access for the client from the owner's existing Android or iOS smartphone (no dedicated hardware purchase). For a single user, a hosting server that will be shared or may be of a low tier and is only expected to keep a database under 1 gigabytes for several years, with the observed booking rate. There is no hardware specification for the barcode or RFID reader because of the reasons given in Section 2.2.5.

Software resources. A mobile browser-based web application, bypassing the hassle of navigating the app store and installation for one user. A relational database management system was chosen for relational integrity of bookings, allocations, labour and cost records. Library for rendering a chart onto a dashboard. A statistical or scripting environment for the seasonal analysis that operates periodically (not in real-time).

People resources. One system user: Business owner. Four workers representing data and wages but without any system access, as per response 11. For the major project, implementation, one developer and one analyst.

Network resources. Client access via mobile data or wi-fi. With the assumption of intermittent connectivity in NFR-05, the design is based on the assumption that there will be no data loss when the entry is booked and the connection is dropped.

Data resources. Master data for equipment categories/variants, owned quantity, unit rental rate and worker records. Transactional data including bookings, allocations, dispatch/return data, labour lines, costs, payments, substitution results and loss and damage data. Availability calculations and per-event profit, as well as monthly demand indices, were derived data.

4.6 USE CASE

MODEL

ACTORS.

- **Owner (primary actor):** the sole system user, who records bookings, checks availability, records costs, labour and returns, and reads the dashboard.
- **Customer (external actor):** initiates a booking enquiry through an external channel; does not interact with the system directly.

- **Analytics component (system actor):** generates demand indices and profit aggregations on a scheduled basis rather than on demand.

USE CASES.

Table 5 : Use Cases

USE CASE	ACTOR	DESCRIPTION	REQUIREMENTS
Check availability	Owner	Query uncommitted quantity per variant across an occupancy window	FR-04, FR-04a
Create booking	Owner	Record a new booking and commit equipment quantities	FR-02, FR-03, FR-05
Negotiate size substitution	Owner	Offer an alternative variant with revised pricing and record the outcome	FR-19
Record dispatch	Owner	Confirm equipment dispatched for an event	FR-06
Record return	Owner	Record equipment returned and flag discrepancies	FR-06
Reconcile loss or damage	Owner	Log missing or damaged items and the charge raised	FR-07, FR-17
Record event cost	Owner	Enter transport, consumable, sub-rental and replacement costs	FR-08
Record labour	Owner	Assign workers to a booking with days, rate and settlement status	FR-20
Record payment	Owner	Record deposit and balance received and payment status	FR-09, FR-18
Generate invoice	Owner	Produce an invoice from booking and pricing data	FR-14, FR-16
View dashboard	Owner	Read booking detail and daily, weekly and monthly profit	FR-10, FR-13
View seasonal analysis	Owner	Read monthly demand indices by category	FR-12
Enter historical booking	Owner	Backfill a past booking from existing records	FR-11
Calculate demand index	Analytics component	Derive monthly demand measures from booking history	FR-12
Aggregate profitability	Analytics component	Compute profit by event type and equipment category	FR-15

Two relationships are modelled. Create booking has an option of Check availability, because availability is checked for in all bookings, not as a separate optional. Negotiate Extension for booking size; Reconcile Extension for loss/damage: each extends the period of time, one under a particular condition, booking size is unavailable, and the other under a particular condition, quantity is not returned as expected.

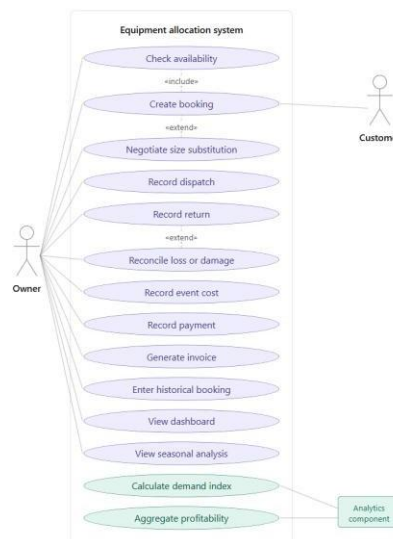


Figure 2 : Use Case Diagram

4.7 DATA FLOW DIAGRAMS

The use case model is used to explain the behavior of the system as seen by its user. In the data flow diagrams below, the movement of data through it is described, which is required in the Report Writing and Formatting Guideline for logical models to be documented for the new system.

Data Flow Diagram (Level 0). The system is depicted as one process communicating with three other external processes. Availability responses and availability reports are provided by the owner, along with the data on the cost and booking of the labour. The

customer is sent an invoice, loss/damage charges and pays with payment records. The substitution path for the replacement of stock and fees created in response 20 creates the partner event company to receive sub-rental requests and the confirmed stock and fees are returned to the partner event company, this flow is because of the substitution path.

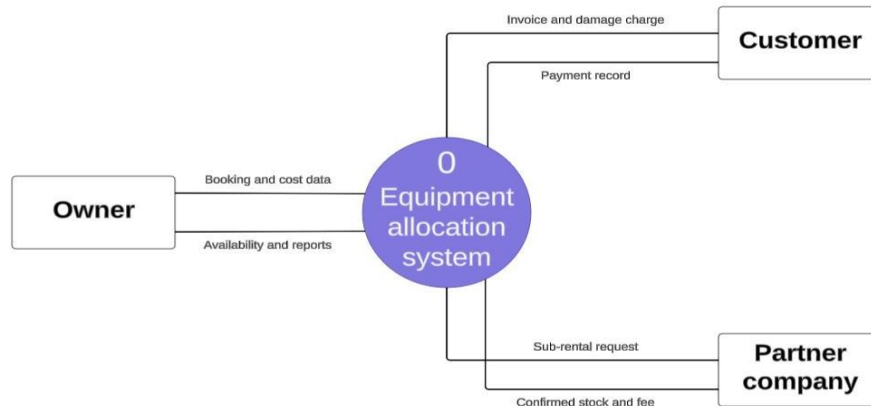


Figure 3 : Data flow diagram, Level 0

Level 1. The single process breaks down into 4 processes: Check availability, Manage bookings, Dispatch and return and Finance and reporting. They have four data stores that support them: D1 Equipment variant, D2 Booking, D3 Allocation, and D4 Finance and damage.

The principal data flows are as follows.

Table 6 : Data Flows

FLOW	DATA
Owner → 1.0	Date and category enquiry
1.0 → Owner	Available quantity per variant
Owner → 2.0	Customer, date, event type, quantities
2.0 → Owner	Quote and confirmation
Owner → 3.0	Dispatched and returned counts

3.0 → Owner	Discrepancy and charge
Owner → 4.0	Cost entries, labour lines and payments
4.0 → Owner	Dashboard, profit, demand index
Partner company → 2.0	Confirmed sub-rental stock and fee
Customer → 4.0	Payment record
4.0 → Customer	Invoice and damage charge

There are three cross-store flows that need to be noted: process 1.0 reads committed quantities from D3 to compute availability; process 2.0 writes allocation records to D3 when a booking is confirmed; process 4.0 reads the booking records from D2 to calculate profit and reports.

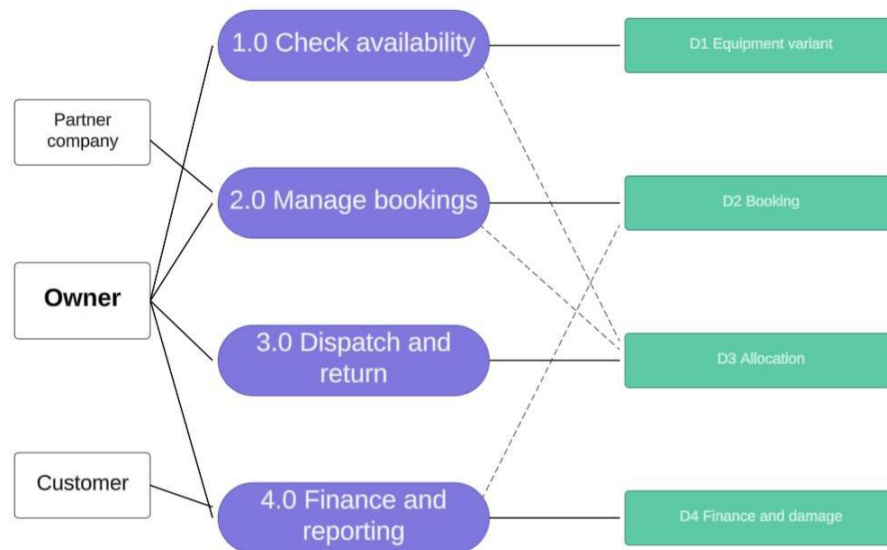


Figure 4 : Data flow diagram, Level 1

4.8 FEASIBILITY ANALYSIS

Technical feasibility. The system does not need to have anything extra to conventional web application development using a relational back-end, and the analytical part employs well-known time series techniques, which can be found in

standard statistical libraries (Borucka, 2023). The most challenging component is determining the occupancy window availability, which requires solving committed quantities over overlapping time windows – a problem that is well-known and that all commercial platforms discussed in Section 2.3 handle well. Technically possible project.

Economic feasibility. The development cost is included as Academic work. Ongoing cost is limited to low tier cloud hosting with one user. Contrary to this, the business does not currently have a way to capture financial information about any event, and so there is no financial evidence to guide decisions about pricing and investment in equipment for an asset array of over one thousand items. Among Malaysian SMEs, Azuar and Nehru (2024) noted that finance is the main constraint for digitalisation, meaning minimal costs of maintenance are necessary. The project has an economic viability.

Operational feasibility. This aspect involves the highest level of risk. Salisu et al. (2021) stated that a majority of business intelligence projects do not achieve the results they had expected and that acceptance by small enterprises is still very low, whereas Wan Mahdi et al. (2025) explain the continued use of manual methods among traditional Malaysian MSMEs as being due to resource constraints and long-standing work practices. There are two points in favour of operational feasibility: the owner has promised to enter each booking personally (response 12), and the system is replacing a manual list rather than an existing digital system, so there is no other tool to take its place. There are also two points against it: the owner lacks any formal technical

training, and the value of the profit reporting is not apparent until after a number of months' worth of data has been accumulated, which results in a period where effort is put in before any benefit is obtained. The project can be considered operationally feasible provided that the interaction cost constraint mentioned in NFR-01 is satisfied.

The schedule is feasible. The scope of the minor project comes to an end at the design stage and can be completed within the semester schedule indicated in Figure 1.1. Implementation will be carried out as part of the major project. The project is therefore schedule feasible.

Legal and ethical feasibility. The system stores customers' names and contact information, records the wages of workers, and holds commercial financial data, which means that it falls within the area covered by Malaysia's Personal Data Protection Act 2010. The obligations are reduced because of the single-user access approach (NFR-07) and since no payment cards or identity documents are stored. The wage records are considered to be employment data and are kept for the owner's own operational purposes and are not disclosed. The owner has given permission for the study to be carried out and for the business to be named. The project is therefore legally and ethically feasible.

4.9 CHAPTER SUMMARY

The system specifications were recorded in this chapter on the basis of requirements collected over three stages. Twenty-one functional requirements and nine non-functional requirements were set up, nineteen of the functional requirements being directly traced to particular stakeholder responses and two being regarded as derived.

Specifications were made for hardware, software, human resources, network resources and data resources, hardware in particular being kept to the smartphone which already belonged to the owner. Logical models were created, including a use case model with three actors and fifteen use cases, a context diagram, and a Level 1 data flow diagram having four processes and four data stores. Feasibility was evaluated in five areas, operational feasibility being found to be the main risk and the sixty-second interaction budget being the condition upon which adoption will depend.

Chapter 5 turns the specifications into a system design, which includes the database structure, the system architecture, the interface design and the method used for demand and profitability reporting.

CHAPTER 5: SYSTEM DESIGN

5.1 INTRODUCTION

The system design set out in this chapter is based on the requirements described in Chapter 4; it includes a specification of the system architecture, the database structure, the allocation logic upon which the assessment of availability relies, the method to be used for demand and profitability reporting, the interface design, and the data entry procedures by means of which the owner will interact with the system.

The two decisions made in Chapter 4 apply to all aspects of that chapter. The first of these is that a booking takes up equipment over an approximate three-day period rather than on a single date (response 13), thereby turning the question of availability into one of overlapping intervals rather than of matching dates. The second point is that the whole design is limited by a data entry budget of about one minute per booking (response 12), this limitation having the same direct effect on the structure of the interface as any other functional requirement.

5.2 SYSTEM ARCHITECTURE

The system adopts a three-tier architecture:

The presentation tier involves the use of a web application that is responsive and accessed via the mobile browser on the owner's current smartphone. Instead of using a native application, a browser-based client is employed so as to avoid the difficulties associated with distribution, installation, and updates for a single user, and thus meet NFR-02 without the need to acquire hardware.

The application level includes server-side logic which is responsible for handling the creation of bookings, carrying out the availability calculation as specified in Section 5.4, computing quotes and costs, as well as generating and authenticating invoices. The analytical routines mentioned in Section 5.5 run on a schedule rather than in response to a request because the demand indices and profitability aggregations change slowly and do not need to be recomputed each time a page is loaded.

The system uses a relational database management system since a relational model has been chosen rather than a document store; the integrity constraints in this area are of a relational type, meaning that an allocation must exist together with both a booking and an equipment variant, a labour line must exist together with both a booking and a worker, committed quantities must be able to be matched against owned quantities, and profit per event depends on joining cost, labour, payment and discrepancy records to a single booking; these are kinds of constraints which the relational engine enforces rather than having to be monitored by an application layer.

Because of NFR-05, the client is built so that it can tolerate intermittent connectivity; bookings are stored in the browser's local state until submission is successful, meaning that if the connection is lost while entering a booking the record will not be lost. And since NFR-06 states that the database is hosted remotely and has scheduled backups, the single point of failure that currently exists because of the owner's phone is eliminated.

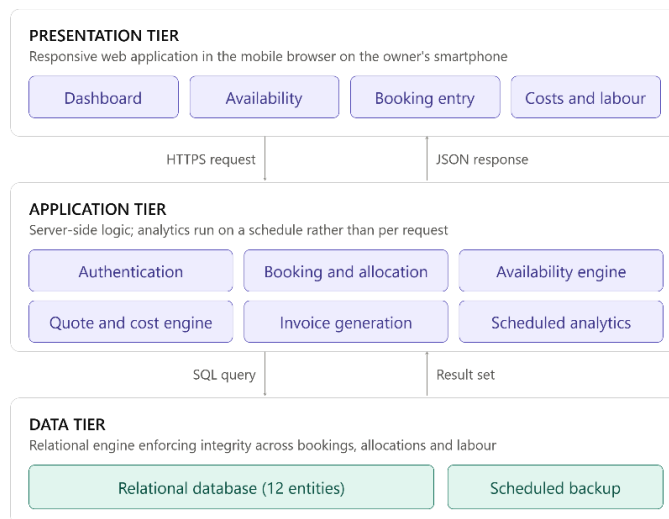


Figure 5 : Three-tier system architecture

5.3 DATABASE DESIGN

5.3.1 ENTITIES AND RATIONALE

Twelve entities are specified. Three design decisions require explanation before the structure is presented.

The category and variant are separated. It has already been established that customers choose colours for tablecloths, chair covers, ribbons, scallops and ceiling drapes. Colour is thus an allocatable dimension and not a descriptive attribute; twenty maroon cloths and twenty gold cloths represent two separate availability pools rather than forty interchangeable items. Similar reasoning applies to canopy size since a 15x20 and a 20x20 are each counted separately. The EQUIPMENT_CATEGORY indicates the general type of item, while the EQUIPMENT_VARIANT contains the specific size, material and colour

combination against which the quantity held is allocated.

Labour is assigned on a per-worker basis rather than on a per-booking basis.

As stated in Response 16, workers are paid RM50 per day, the amount increasing to RM70 for more demanding work; this alone would have allowed the labour cost to be stored as a single computed figure for each booking. However, according to Response 22, wages are paid individually and may remain unpaid after a job has been completed. Since a single total figure could not show which worker had been paid and which had not, labour is represented by a WORKER entity which contains each worker's identity and their default rate, and by a LABOUR_LINE associative entity which includes one record for each worker per booking, specifying the number of days worked, the applicable rate and the settlement status.

The results of substitutions are recorded rather than just provided. As Response 20 showed, when a size is unavailable an alternative is offered, which the customer then accepts or rejects, and in some cases orders that are rejected are fulfilled using stock that has been sub-rented from a partner company. The SUBSTITUTION_LOG records the outcome so that the sub-rental costs can be assigned to the booking which caused them.

Table 7 : Entities And Rationale

ENTITY	PURPOSE
CUSTOMER	Identity and contact details of the booking party
BOOKING	One event: dates, occupancy window, event type, source channel, quote, status
EQUIPMENT_CATEGORY	General class of item (canopy, table, chair, cloth, fan)
EQUIPMENT_VARIANT	Specific size, material and colour combination, with owned quantity and replacement costs

ALLOCATION	Quantity of one variant committed to one booking, with dispatched and returned counts
DISCREPANCY	Loss or damage identified at return, with the charge raised
EVENT_COST	Non-labour costs incurred against a booking, including sub-rental
WORKER	Each worker's identity, default daily rate and active status
LABOUR_LINE	One worker's engagement on one booking: days, rate, amount and settlement status
PAYMENT	Deposit and balance payments received
SUBSTITUTION_LOG	Record of a substitution offer and its outcome
RATE_CARD	Effective-dated unit rates per variant

5.3.2 RELATIONSHIPS AND CARDINALITY

Table 8 : Relationships And Cardinality

RELATIONSHIP	CARDINALITY	JUSTIFICATION
CUSTOMER to BOOKING	One to many	Referral and repeat business means one customer may book repeatedly
BOOKING to ALLOCATION	One to many	A single event commits several equipment variants
EQUIPMENT_VARIANT to ALLOCATION	One to many	A variant is committed across many bookings over time
EQUIPMENT_CATEGORY to EQUIPMENT_VARIANT	One to many	A category holds several size and colour variants
ALLOCATION to DISCREPANCY	One to zero or many	Most allocations return complete; some produce loss or damage
BOOKING to EVENT_COST	One to many	Transport, consumables and sub-rental recorded separately
BOOKING to LABOUR_LINE	One to many	Several workers are engaged on each event
WORKER to LABOUR_LINE	One to many	A worker is engaged across many bookings over time
BOOKING to PAYMENT	One to many	Deposit and balance are separate payments
BOOKING to SUBSTITUTION_LOG	One to zero or many	Substitution occurs only when a preferred variant is unavailable
EQUIPMENT_VARIANT to RATE_CARD	One to many	Rates change over time; history is retained

Many-to-many relationships are handled by means of associative entities. The

relationship between bookings and equipment variants is dealt with by means of ALLOCATION, which includes the committed, dispatched and returned quantities itself. The relationship between bookings and workers is managed by means of LABOUR_LINE, which contains the days, the rate and the settlement status. In neither case is it a simple junction table since in both instances the entities hold attributes that belong to the association rather than to either of the participants.

There is a uniqueness constraint on LABOUR_LINE for the combination (booking_id, worker_id) because a worker is engaged on a particular booking only once, with the number of days being recorded as a quantity not as separate rows.

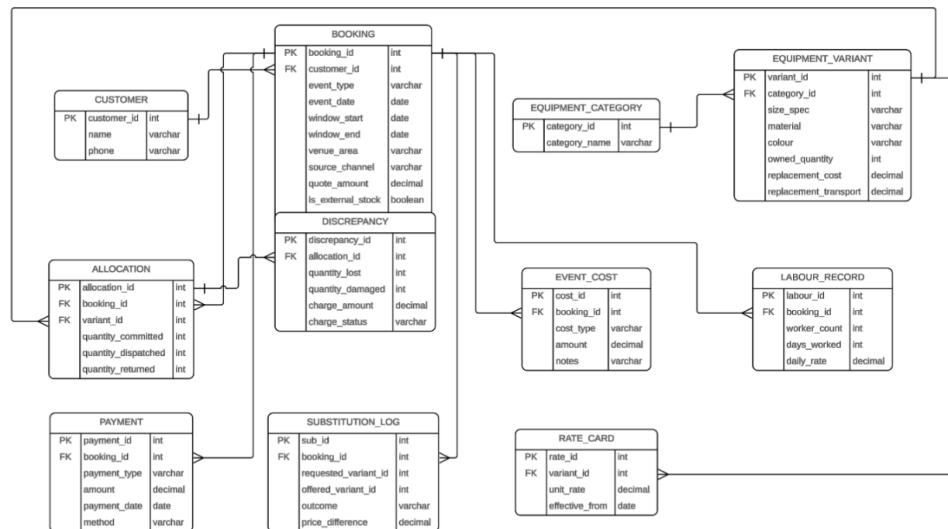


Figure 6 : Entity-relationship diagram

5.3.3 DATA DICTIONARY — PRINCIPAL TABLES

BOOKING

Table 9 : Booking

ATTRIBUTE	TYPE	NOTES
booking_id	Integer	Primary key
customer_id	Integer	Foreign key to CUSTOMER
event_type	Enumerated	Wedding, kenduri, corporate, anniversary, birthday, Nichayathartham
event_date	Date	The day of the event itself
window_start	Date	Defaults to event_date minus one day; overridable (FR-04a)
window_end	Date	Defaults to event_date plus one day; overridable
venue_area	Text	Used to distinguish within-Selangor delivery from outside
source_channel	Enumerated	Instagram, WhatsApp, phone, walk-in, referral
quote_amount	Decimal	Derived from RATE_CARD and allocated quantities
is_external_stock	Boolean	True where stock was sub-rented from a partner company
status	Enumerated	Enquiry, confirmed, dispatched, returned, closed, cancelled

EQUIPMENT_VARIANT

Table 10 : Equipment_Variant

ATTRIBUTE	TYPE	NOTES
variant_id	Integer	Primary key
category_id	Integer	Foreign key to EQUIPMENT_CATEGORY
size_spec	Text	For example 20x20, or null where not applicable
material	Text	Transparent, normal, fibre, wood, plastic, banquet, fabric
colour	Text	Null where colour is not selectable
owned_quantity	Integer	Total units owned
replacement_cost	Decimal	Market price of a replacement unit
replacement_transport	Decimal	Transport cost of obtaining a replacement (response 19)

WORKER

Table 11 : Worker

ATTRIBUTE	TYPE	NOTES
worker_id	Integer	Primary key
name	Text	Worker identity, held for the owner's operational use
default_daily_rate	Decimal	Normally RM50; overridable per engagement
is_active	Boolean	Distinguishes current from former workers

LABOUR_LINE

Table 12 : Labour_Line

ATTRIBUTE	TYPE	NOTES
labour_line_id	Integer	Primary key
booking_id	Integer	Foreign key to BOOKING
worker_id	Integer	Foreign key to WORKER
days_worked	Integer	Defaults to the occupancy window length
daily_rate	Decimal	RM50 standard or RM70 for heavier work (response 16)
amount_earned	Decimal	Derived as days_worked multiplied by daily_rate
payment_status	Enumerated	Pending or paid
paid_date	Date	Null while pending

The charges are computed as discrepancy, defined as charge.amount , with its computation being $\text{quantity} \times \text{replacement_cost} + \text{replacement_transport}$, which is the request by the owner of the rule. The charge is computed as discrepancy, charge.amount ; directly as per the rule the owner described, a chair priced for RM5 is charged at RM7.

5.3.4 OPEN DESIGN ITEM: PRICING STRUCTURE

The owner acknowledged that there is a price list, that the price starts at RM250 and increases with the number of canoppes, tables and chairs sold, and that delivery to Selangor is free (responses 14 and 15). It is not given whether the price

list document is a per unit rate across the allocated items, or a series of price increments for a defined package set with additions incrementing the price, one structural question has yet to be resolved.

They both fit in the design. RATE_* are unit rates per variant of the summation model. If the price list is tier-based, then a PACKAGE_TIER entity is created containing tier name, base price and the quantities included in the tier, and BOOKING.quote_amount would be the tier matched base price plus additional per-unit rates for any items not included in the matched tier. This choice is indicated by the open item recorded and is considered prior to the start of the major project. There's nothing to assume; if the system wrongly guessed on one pricing model, all the profits the system produces will be oddly rendered invalid.

As regarding response 15, a follow-up observation can be drawn out and developed in the following Chapter: Free delivery in Selangor is actually a cost that is not paid, never charged, nor recorded, and which therefore affects the margin in an invisible way on every job. The first step towards getting it visible is to capture it as an EVENT_COST of type transport.

5.4 ALLOCATION LOGIC DESIGN

Availability determination is the technically substantive part of this design and warrants explicit specification.

Because each booking occupies stock from window_start to window_end inclusive,

two bookings compete for the same variant whenever their windows overlap, even if their event dates differ. An event on 14 June occupies stock from 13 to 15 June; an event on 16 June occupies stock from 15 to 17 June. The two overlap on 15 June and therefore cannot jointly commit more than the owned quantity of any variant.

The technically substantive component of this design is determinations of availability, determinations determine the functioning of the device.

```
FUNCTION available_quantity(variant_id, window_start, window_end):  
  
    owned = EQUIPMENT_VARIANT.owned_quantity WHERE variant_id matches  
  
    peak_committed = 0  
  
    FOR each day d FROM window_start TO window_end:  
  
        committed_on_d =  
  
        SUM(ALLOCATION.quantity_committed)  
  
            WHERE ALLOCATION.variant_id matches  
  
            AND ALLOCATION.booking.status IN (confirmed,  
  
            dispatched) AND d BETWEEN booking.window_start  
  
            AND booking.window_end  
  
        IF committed_on_d > peak_committed:  
  
            peak_committed = committed_on_d  
  
    RETURN owned - peak_committed
```

The average is not taken instead the peak is, since if a variant is fully committed on even a single day within the requested time period then it becomes unavailable. 62

Cancelled and enquiry-status bookings are omitted in order that unconfirmed enquiries may not prevent genuine availability from being confirmed.

The FR-05 rule works by rejecting any request for an allocation that exceeds this amount; since Response 21 has shown that partial fulfilment has never taken place, allocation is considered on an all-or-nothing basis at the variant level and not allowing for a shortfall to be accepted.

Substitution logic: if the desired variant is not available, the FR-19 system must identify possible substitutes. By following the procedure outlined in response 20, the system looks for variants in the same category that have a specification equal to or greater than that of the one requested and which are sufficiently available over the time window in question, then displays these options together with the price difference calculated from the RATE_CARD. The system itself does not make the decision to substitute; it merely offers the option, and the result whether the substitute is accepted, rejected, or fulfilled using externally rented stock is entered into the SUBSTITUTION_LOG. In cases where external stock is used, the BOOKING.is_external_stock flag is set and the sub-rental fee is recorded as an EVENT_COST, since otherwise those bookings would show inflated profits.

5.5 ANALYTICAL DESIGN

5.5.1 PROFIT COMPUTATION

Profit for a booking is computed as:

$$\text{revenue} = \text{quote_amount} + \text{SUM}(\text{DISCREPANCY.charge_amount})$$

$$\begin{aligned}
\text{labour_cost} &= \text{SUM}(\text{LABOUR_LINE.days_worked} \times \\
&\text{LABOUR_LINE.daily_rate}) \text{ costs} = \text{labour_cost} + \\
&\text{SUM}(\text{EVENT_COST.amount}) \\
\text{profit} &= \text{revenue} - \text{costs}
\end{aligned}$$

labour_cost is the sum of the booking's labour lines rather than a stored aggregate, which follows from the per-worker structure established in Section 5.3.1. days_worked defaults to three to reflect the occupancy window and daily_rate defaults to each worker's default_daily_rate, with an override to RM70 for heavier work.

Transport is the one estimated component, since the business uses its own lorry and Hilux (response 17) and therefore incurs fuel and maintenance rather than an invoiced charge. The design records a per-trip transport estimate as an EVENT_COST, and Chapter 6 notes this as the least precise figure in the model. Aggregation by event type and by equipment category, required by FR-15, is performed over these per-booking figures.

5.5.2 SEASONAL DEMAND ANALYSIS

Following Tadayonrad and Ndiaye (2023), seasonality is expressed as an index rather than a description. For each equipment category and calendar month:

$$\begin{aligned}
\text{demand_index}(\text{category}, \text{month}) &= \text{mean units committed in that month} \\
&\div \text{mean units committed across all months}
\end{aligned}$$

An index above 1.0 identifies a month of above-average demand for that category, and the resulting profile can be compared against the owner's stated impression that February, April, May, June, August, September and December are busy while January, March, October and November are slower (response 9).

Borucka (2023) demonstrates that seasonal ARIMA, exponential smoothing and TBATS methods can be applied effectively by firms possessing only a series of historical sales observations, and that increased model complexity does not reliably improve accuracy for seasonal series. On this basis, exponential smoothing with a seasonal component is specified as preferred over the machine learning approach, not only with respect to accuracy but also because of the interpretability.

A necessary qualification. These techniques need a long series of monthly observations, which are usually at least two full seasons to obtain a reliable estimate of the seasonal component. There was 1 invoice there and 1-2 months of records at the time of writing, that the owner was able to provide. Forecasting thus is designated as a capability, not as available at launch, as records are built over the years. Until then, the system calculates and plots the monthly demand taken from observations, but does not predict the monthly demand based on the system; the owner's stated seasonal patterns are used as assumptions for testing which may not be the final conclusion. Giving away a prediction based on two months' worth of information would be worse than no information, since the forecaster would be assuming an aura of authority that is not due. This is made very clear on the UI, which uses the term "observed" for the seasonal screens indicating that there is no forecast

5.6 INTERFACE DESIGN

Nine screens are specified. Their aim is to enable singlehand operation with a phone because the owner uses his mobile phone as his tool and is often deployed while performing other tasks. All values presented in the figures are indicative not actual business values.

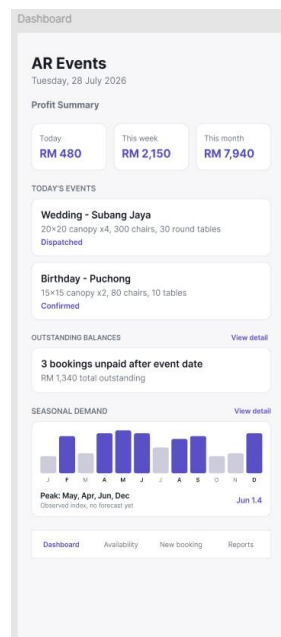


Figure 7 : Dashboard

Default landing view: day, week and month profit; today's events, outstanding balances, compact seasonal index.

Availability check

Check availability
Answer a customer without hanging up

EVENT DATE
Sunday, 14 June 2026
Stock held 13 Jun to 15 Jun (auto)

Canopy 20×20 normal	14 free of 19
Canopy 15×20 normal	0 free of 3
Canopy 15×15 normal	2 free of 4
Plastic chair	540 free of 700
Banquet chair	100 free of 100
Round table	31 free of 50
Air cooler	4 free of 7
Extension cable	12 free of 20

15×20 not available on this window

Figure 8 : Availability check

Answer a customer mid-conversation, showing the derived window and substitution prompt.

New booking

New booking
Defaults pre-filled. Change only what differs.

EVENT DATE
14 Jun 2026

EVENT TYPE
Wedding

CUSTOMER
Kumar R.

CHANNEL
WhatsApp

Canopy 20×20	14 free	- 4 +
Round table	31 free	- 30 +
Plastic chair	540 free	- 300 +
Table cloth - maroon	44 free	- 30 +
Stand fan	10 free	- 6 +

Quote from price list
Deposit due: RM 640

RM 1,280

Save booking

Figure 9 : New Booking

Create a booking within the entry budget using pre-filled defaults and quantity steppers.

Record return

Record return

Wedding - Subang Jaya, 14 Jun 2026

Returned counts pre-filled to dispatched. Change only shortfalls.

Canopy 20x20 normal Sent 4	4 back
Round table Sent 30	30 back
Plastic chair Sent 200	296 back 4 lost
Table cloth - maroon Sent 30	28 back 2 damaged
Stand fan Sent 6	6 back

CHARGE TO CUSTOMER (AUTO)

Plastic chair x4 RM 5 + RM 2 transport	RM 28
Table cloth x2 RM 25 + RM 2 transport	RM 54
Total charge	RM 82

Confirm return and raise charge

Figure 10 : Record Return

Record what came back and raise the calculated loss or damage charge.

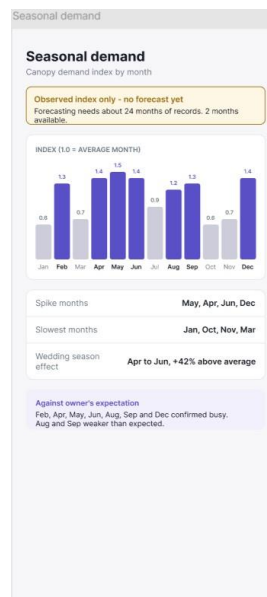


Figure 11 : Seasonal demand

Full monthly demand index with cross-check against the owner's expectation.

Add equipment item

Add equipment item
Equipment register

CATEGORY
Table cloth

SIZE
Not applicable

MATERIAL
Fabric

COLOUR
Maroon

Each colour is counted separately. Maroon cloths cannot fill an order for gold.

OWNED QUANTITY
46

RENTAL RATE
RM 8

MARKET PRICE
RM 25

REPLACEMENT TRANSPORT
RM 2

Loss or damage charge
RM 25 + RM 2, calculated **RM 27**

ALREADY IN THIS CATEGORY

● Gold	40
● White	60
● Maroon	46 (new)

Add to register

Figure 12 : Add Equipment Item

Register a variant with owned quantity, rate, market price and replacement transport.

Event costs

Event costs
Wedding - Subang Jaya, 14 Jun 2026

OTHER COSTS	
Labour 4 workers x 3 days - see labour record	RM 600
Transport Owning and Hired, fuel estimated	RM 120
Consumables Cable ties, tape	RM 45
Sub-rental No external stock used	RM 0
REVENUE	
Quote from price list	RM 1,280
Loss and damage charge 4 chairs, 2 cloths	RM 82
PAYMENT	
Deposit received 2 Jun 2026, term transfer	RM 640
Balance outstanding Overdue since 14 Jun	RM 722
Profit for this event RM 1,362 revenue less RM 765 cost	RM 597

Figure 13 : Event Costs

Record transport, consumables and sub-rental; view revenue, payment and per-event profit.

Labour record

Labour record

Wedding - Subang Jaya, 14 Jun 2026

Total

RM 600

Paid

RM 450

Pending

RM 150

3 days worked: setup 13 Jun, event 14 Jun, teardown 15 Jun

WORKERS ON THIS JOB

Worker A

3 days x RM 50

RM 150

Paid

Worker B

3 days x RM 50

RM 150

Paid

Worker C

3 days x RM 50

RM 150

Paid

Worker D

3 days x RM 50

RM 150

Pending

Rate RM 50 per day, RM 70 for heavy work

Mark Worker D as paid

Figure 14 : Labour Record

View each worker’s days, wage and settlement status for a booking.

Add labour

Add labour

Wedding - Subang Jaya, 14 Jun 2026

SELECT WORKER

Worker A

Standard RM 50

On job

Worker B

Standard RM 50

On job

Worker C

Standard RM 50

On job

Worker D

Standard RM 50

Worker E

Casual hire RM 50

+ Add a new worker

DAYS WORKED

- 3 +

Window is 3 days

DAILY RATE

RM 50

RM 70

RM 70 for heavy work

PAYMENT STATUS

Pending

Paid

Wage for Worker D

3 days x RM 50, calculated

RM 150

Add labour line

Figure 15 : Add Labour

Assign a worker to a booking with days, rate and settlement status.

These screens are based on two design principles, both of which come from Chapter 2. In the way that Gonçalves et al. (2023) suggest, the dashboard focuses on reading speed rather than on analytical depth, since the three profit figures, the day's events and the seasonal strip can be seen without having to make any settings, otherwise a dashboard which needs to be set up before it can provide any information would not be checked on a daily basis. As Alsibhawi et al. (2023) have stated in their framework where perceived ease of adoption and information quality are identified as factors that affect the use of business intelligence in small enterprises each screen includes defaults instead of leaving fields empty whenever a sensible default is available.

Various screens show the specific requirements clearly. The availability screen shows the derived occupancy period rather than just the event date and also presents a zero-availability option together with a substitution proposal including the price difference, thus fulfilling FR-04a, FR-05 and FR-19. When the add equipment screen records the market price and the replacement transport at the time of registration, the return reconciliation screen is then able to calculate the charge without the owner having to do anything, thus demonstrating FR-01 feeding into FR-17. The add labour screen disables any workers who have already been assigned to the booking, making the uniqueness constraint mentioned in Section 5.3.2 evident.

The interface language is English (NFR-09).

5.7 DIALOGUE AND DATA ENTRY PROCEDURES

The one-minute entry budget is met through the following procedures.

Values are derived rather than inputted. The occupancy window is calculated on the basis of the event date rather than being entered, and an override is provided. The quote is calculated using the allocated quantities and the rate card. Labour days default to the length of the window and the rate defaults to that of each individual worker. Discrepancy charges are calculated on the basis of the replacement cost together with replacement transport. In all these instances the owner confirms or overrides rather than carrying out the calculation.

The process is progressive. A booking can be saved at enquiry status even if the only information recorded is the date, the name of the customer, and the quantities of the equipment, the cost, the labour details and the payment information being added at a later stage. This is important since the time at which the booking is made is a live conversation whereas the time when the cost is recorded is not; if both aspects were required at the same time, it would ensure that neither would be carried out.

The input controls are appropriate for the task; for the quantity fields, steppers are used instead of allowing free numeric entry since it is faster to enter a chair booking of 300 using a stepper preset than by typing it. The daily rate employs a two-option control rather than a numeric field because only two rates are available and two taps cannot result in a typo. For dates, a calendar picker is used with the default set to the next weekend since most events take place on weekends.

Validation involves checking the committed quantity against the available quantity at the time the entry is made, not when the entry is submitted, so that the owner becomes aware of any discrepancy while still talking to the customer. The quantity that is returned must not be greater than the quantity that has been dispatched. A worker cannot be included twice in the same booking. A balance may not be stated as settled if the payments recorded are less than the quote, this dealing with the late payment pattern mentioned in response 18.

Dialogues requiring confirmation are only needed in cases where the action cannot easily be undone, such as cancelling a previously confirmed booking, bringing a damage charge against a customer, and marking a wage as paid. Routine saves take place without any interruption, since it would occupy a significant portion of the entry budget if a confirmation dialogue were displayed for every action.

5.8 CHAPTER SUMMARY

This chapter laid down a three-tier architecture which included a browser-based mobile client, a relational database containing twelve entities that separated the equipment category from the allocatable variant so that colour and size could each act as independent availability pools, and included the modelling of labour on a per-worker basis in order that individual wage settlements could be tracked. The allocation algorithm calculated availability as the owned quantity minus the peak committed quantity over the occupancy period, and the substitution logic enabled the negotiation process as described by the owner, including the recording of externally sub-letted

stock as a cost. Profit computation obtained labour from the total of the labour lines associated with a booking, transport being identified as the only estimated component. Seasonal analysis was set out as a monthly demand index with forecasting being postponed until a sufficient number of observations had accumulated. Nine screens together with a series of data entry procedures were designed in accordance with the one-minute entry budget, and the pricing structure was listed as an open design item until the price list document was available.

Chapter 6 presents the results of the project, assesses the framework in relation to the objectives established in Chapter 1, identifies its limitations, and suggests the kinds of work that should be carried out in the major project.

CHAPTER 6: CONCLUSION

6.1 INTRODUCTION

The minor project is brought to a close in this chapter, with a summary of the work carried out, an assessment of the framework in relation to the four objectives set out in Section 1.5, a frank statement of the design's strengths and weaknesses, recommendations for the kind of work to be done in the major project, and a reflection on the development process and the lessons it has provided.

6.2 SUMMARY OF PROJECT OUTCOMES

The project resulted in a written design of an integrated real-time system for allocating tents and equipment for AR Events, which is an event equipment rental company based in Bandar Sunway, Petaling Jaya, and looks after more than one thousand rentable assets by means of entries on a mobile phone list.

The following are the deliverables: a problem definition based on a structured interview with the business owner rather than on assumptions; a literature review covering fifteen sources published between 2020 and 2026, the review covering the topics of inventory management in small enterprises, decision support for SME operations, seasonal demand forecasting in the face of data scarcity, business intelligence adoption, and asset tracking technologies, together with a comparative analysis of six commercial rental management platforms; a documented methodology that selects the Waterfall model together with the reasons for this choice and the

limitations associated with it; a system analysis including twenty-one functional requirements, nineteen of which are traced to specific statements from stakeholders, nine non-functional requirements, full resource specifications, and a use A system design consisting of a three-tier architecture, a twelve-entity relational database schema, an allocation algorithm which resolves availability during overlapping occupancy periods, a specified method for calculating per-event profit and monthly demand indices, and nine interface screens designed with a one-minute data entry budget.

The most important finding of the project is not a design choice but an operational fact: the company is unable to state the profit made from any particular event, even though all the elements needed to calculate it are already contained in the records which the company produces and throws away. The cost of labour is known since it is a fixed daily rate over a known number of days. Revenue is determined by a documented price list. Loss recovery is governed by a clear rule. What is missing is not information but rather structure.

6.3 ACHIEVEMENT OF OBJECTIVES

Objective 1, which is to analyse existing practices, has now been achieved. For the first time in the company's history, the booking, allocation, recovery, labour and record-keeping procedure at AR Events is set out in Section 4.2, covering twenty-two areas of inquiry and with each further requirement being traceable to a particular response.

Objective 2 has been achieved, which is to design a real-time allocation model. In section 5.4 an algorithm is specified for calculating the uncommitted quantity per variant as the owned quantity minus the peak committed quantity over the booking's occupancy window. The specification properly addresses the case which sets this domain apart from ordinary inventory that of two events on consecutive days competing for the same stock since their three-day windows overlap.

Objective 3 has been achieved, which is to design a data structure that records the cost and revenue for each event. Section 5.3 identifies twelve entities that allow the per-event profit to be calculated from revenue, labour costs, event costs and discrepancy charges, with the option of aggregating the data by event type and equipment category.

Objective 4 namely, to put forward an analytical approach and dashboard has been only partly achieved, and the gap in this is clearly stated. The design of the dashboard and the method used to calculate the monthly demand index are fully detailed and illustrated in Figures 5.3 and 5.7. The forecasting aspect has been specified but not demonstrated, since only one invoice was available when the text was written, whereas the methods in question require two complete seasonal cycles. Although Objective 4 was phrased as involving the identification and quantification of seasonal patterns rather than prediction, it has therefore been met on that basis; however, if one considers the wider goal of demand prediction, then it has only been specified and not actually delivered.

6.4 SYSTEM STRENGTHS

The design is based on an actual operation; each requirement can be traced back to a statement made by the business owner, and two of the requirements were altered when his answers contradicted previous assumptions. Rather than hiding them, Section 4.2 lists both corrections.

The allocation method deals with the real issue; the key element of the design consists in treating availability as an interval-overlap problem rather than a date-matching one, a approach which is directly based on an operational fact the three-day occupancy window that a general inventory design would have overlooked.

The construction of the cost baseline is taken to be the system's first function. As section 2.3 showed, all of the six commercial platforms under review assume that the operator already knows their unit costs. This framework contradicts that assumption by taking account of the market price and replacement transport at the time of registration and labour at the time of engagement, which is the particular gap that it addresses.

The design is tailored to the user's needs. It is based on a single non-technical user, using a browser instead of an app, providing default values rather than blank fields, employing steppers instead of numeric keyboards, and setting a sixty-second interaction budget based on the actual commitment made by the owner. As Salisu et al. (2021) discovered, most business intelligence initiatives fail to deliver on their promises and adoption within small enterprises stays low; the only available defence

against this result is to design according to a stated tolerance rather than to a feature list.

Limits are not concealed but stated outright; for example, the unresolved pricing structure, the estimated transport figure, and the deferred forecasting are each mentioned in the chapter in which they appear.

6.5 SYSTEM WEAKNESSES AND LIMITATIONS

The cost of transport is estimated rather than actually measured. Since the business owns the lorry and the Hilux there are no invoices for each journey; the fuel and maintenance costs have to be shared. This is the least accurate figure in the profit calculation, and any profit figure calculated on a per-event basis inherits this lack of accuracy.

The business does not take into account the cost of free delivery within Selangor. As established in Response 15, delivery is given free of charge in Selangor and is never charged as a separate item. It is still a genuine cost in terms of fuel, the deterioration of vehicles, and the time taken by workers, and since it is not recorded, it slowly reduces the margin on each job without being visible. The framework is able to make this cost visible by treating it as a transport expense, but it cannot decide whether the present It is a commercially sound practice. The question in issue whether the benefit to customer acquisition from free delivery is worth the cost, or whether it is simply an unexamined habit that needs the data which this system would produce must be addressed.

The pricing model has not yet been decided upon. At the time of writing the price list was not available. Although the design allows for either a per-unit summation model or a package tier model, the decision has to be made before implementation, and if the wrong option is chosen then all the quotes and profit figures already produced would be invalidated.

Forecasting has been specified but is not operational. The available historical records cover a period of only one to two months and therefore cannot be used for seasonal forecasting. Until approximately two years' worth of bookings have been accumulated, the system displays the actual demand rather than the forecasted demand.

The findings cannot be applied more broadly since this study is based on a single case. The framework applies to a business that has this asset profile, this staffing model and this booking pattern; it has not been tested against event rental businesses as a whole.

The Waterfall model revealed its well-known weakness on two occasions: the occupancy window requirement led to a need for revising the documented requirements, and the wage settlement requirement only came to light during interface design. An iterative methodology could have picked up both of these issues earlier.

It has been deliberately arranged so that workers are excluded, which in turn affects the quality of the data although the workers identify losses and damage when disassembling the equipment, it is the owner who records them afterwards and thus the record relies on accurate communication. Response 11 found that workers do not need access to the system and this requirement has been honoured, but this does place

limit on the accuracy of the reconciliation of returns.

No tests have been carried out; in keeping with the small project scope, testing, user acceptance evaluation and implementation are mentioned in Section 3.6 but have not actually been carried out.

6.6 RECOMMENDATIONS FOR FUTURE ENHANCEMENT

For the major project, in order of priority get the price list and organise it, settling the open issue in Section 5.3.4 carry out the allocation algorithm and test it using the boundary conditions mentioned in Section 3.6, with a special focus on consecutive-day bookings competing for a single variant; backfill the historical bookings using the invoices, chat histories and bank transfer records referred to in response 8, as this will both fill in the demand index and check the schema against actual data; and construct the nine screens before carrying out the user acceptance evaluation with the owner in the way described in Section 3.6.

Beyond the main project, the forecasting component should be enabled when enough records have accumulated, employing the exponential smoothing method described in Section 5.5.2. If the owner's attitude towards worker access changes, a limited return-confirmation function for workers should be introduced, thereby tackling the data quality ceiling mentioned above. The cost of free delivery should be determined once transport data is available, so that the practice can be assessed rather than merely assumed. Maintenance scheduling based on the demand index should be considered, with repairs and replacements for items that have high loss rates being carried out

during the slower months. Although asset tagging is still not advisable for chairs and tables for the reasons stated in Section 2.2.5, it might prove worthwhile for the more valuable canopy sets alone.

6.7 REFLECTION ON THE DEVELOPMENT PROCESS AND LESSONS LEARNED

What you ask about is revealed by the interviews. The key result of the project, the three-day occupancy window, was the answer to a follow-up question which had only been asked since the initial version of the availability requirement seemed too simple; if that question never had been asked, the design would have compared availability with a single date and would have been wrong in a way that could not have been corrected even with the greatest care in implementation.

Designing an interface puts the data model to the test. The need for settling each worker's wage did not come about as a result of the interview; it arose when we tried to draw a labour screen and realized that a single total figure was not sufficient to show which worker had been paid. Creating a representation of an object is a more rigorous test of understanding than simply describing it.

Being already familiar with the situation is a advantage as well as a disadvantage. The fact that we had previous experience with AR Events greatly reduced the time taken for the elicitation process, but it also led to two wrong assumptions regarding availability, and these errors were only identified because they were specifically raised with the owner rather than being accepted on trust.

An obvious gap is more compelling than one that is hidden. In this report there are three unresolved issues: pricing structure, transport precision, and forecast viability. These issues can be cited individually when they arise, rather than creating a design that seems complete but doesn't meet the price list.

Simple was not an end goal, it was a destination. According to Borucka (2023), increased model complexity does not reliably lead to better accuracy in seasonal forecasts, and this finding meant that it was reasonable to choose interpretable exponential smoothing rather than a machine learning method. Tan and Sidhu (2022) explained the purpose of asset tagging, which in turn provided a reason to reject it when dealing with an inventory of eight hundred outdoor-handled chairs. In each instance the literature supported the simpler option, and that simpler option was the one that the user could actually put into practice.

6.8 CONCLUSION

The aim of this project was to provide a manually operated event rental business with visibility of its equipment on an asset basis and with respect to the quantity of its profits. It includes a documented, traceable, and internally consistent design of that framework, along with the interface and data models needed to put it into effect.

The framework doesn't specify to AR Events how it should run its business instead, it makes the business understandable to its owner by showing which stocks are committed on which days, what each event costs, during which months there are genuine spikes, and which wages have not been paid. The fact that a business has

reached a size of over one thousand assets merely by recalling them proves that recollection functions. This approach has been adopted since recollection cannot scale, and because the questions the owner is currently unable to answer are exactly those that decide whether the next one thousand assets will be purchased wisely.

The implementation, testing and evaluation come next with respect to the main project, for which this report serves as the specification.

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